

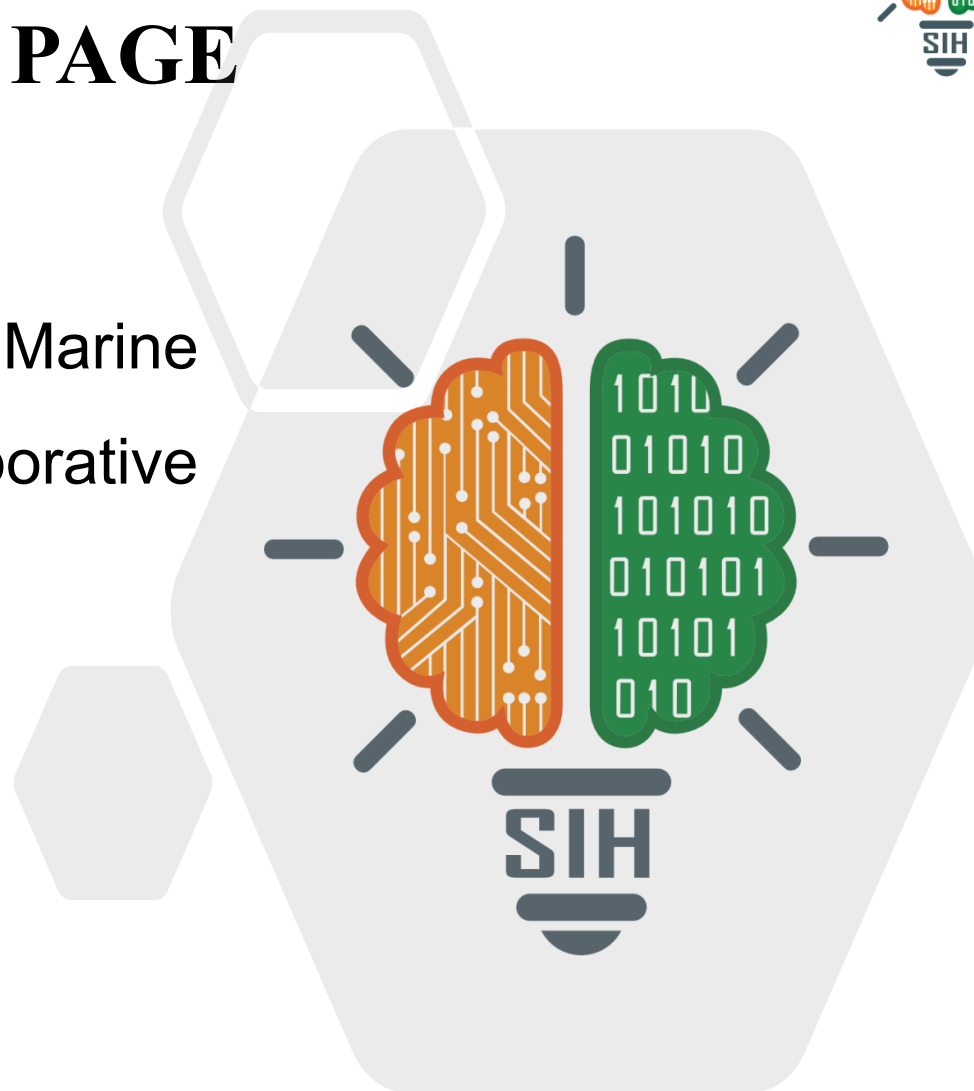
SMART INDIA HACKATHON 2026

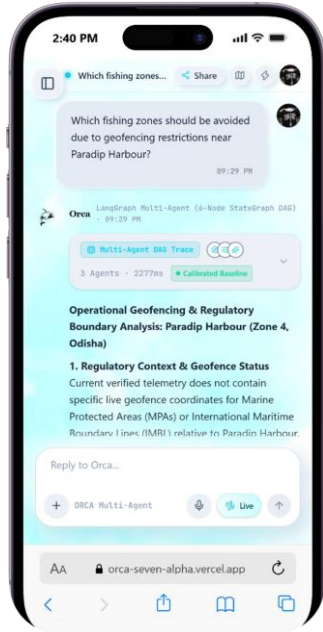


SMART INDIA
HACKATHON
2026

TITLE PAGE

- **Problem Statement ID – SIH26176**
- **Problem Statement Title-** ORCA Marine EcOsysteM Reasoning with Collaborative Agents
- **Theme-** Space Technology
- **PS Category-** Software
- **Team ID-**





Live ORCA conversational interface

THE CHALLENGE

- **Massive data volume:** ISRO generates vast satellite EO data daily — SST, chlorophyll, currents, forecasts.
- **Accessibility gap:** Fishermen, coastal authorities & researchers can't easily interpret complex marine datasets.
- **No intelligent interface:** No conversational platform exists for spatial-temporal reasoning over heterogeneous data.
- **Language barrier:** End-users speak regional Indian languages, not technical English.

ORCA SOLUTION

- **Agentic AI platform:** Multi-agent system — Planner, Ocean, Weather & Risk specialists collaborate autonomously.
- **Natural language:** Users ask in any Indian language and get evidence-backed, plain-language answers.
- **Spatial-temporal reasoning:** Autonomous data discovery across MOSDAC, INCOIS ERDDAP & GIS boundary datasets.
- **Multimodal output:** Text explanations, interactive maps, zone markers, alerts & cited evidence trails.

8 CORE USER QUERIES ORCA HANDLES

PFZ Zones

Where is the nearest Potential Fishing Zone today?

Sea Safety

Is it safe to venture into the sea tomorrow morning?

Tide & Weather

What are the tide, weather & sea conditions near my location?

Cyclone Alerts

Are there any lightning or cyclone alerts in my area?

SST & Chlorophyll

Which regions show high chlorophyll & favorable sea surface temp?

Route Planning

What is the safest route given current weather and sea state?

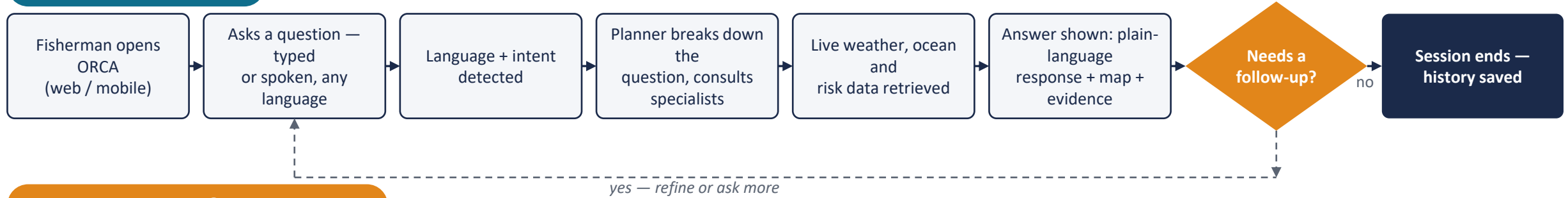
Fish Decline

Why has fish productivity declined in a particular coastal region?

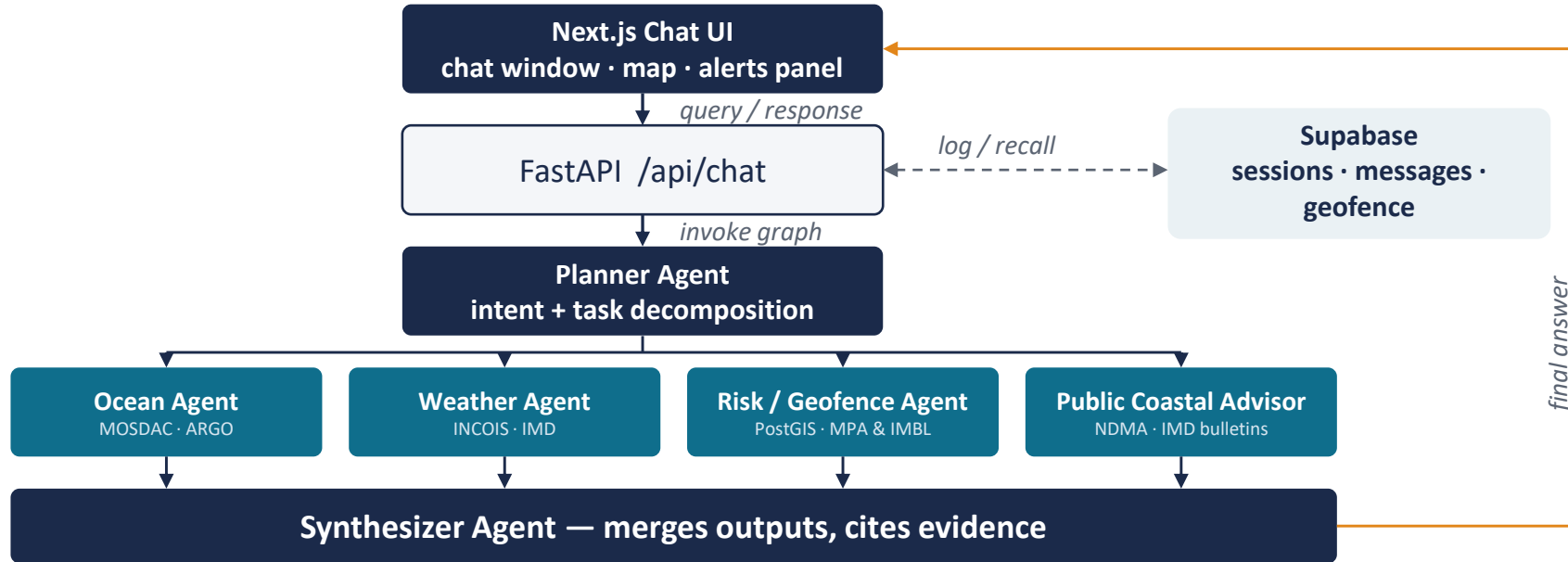
Geofencing

Which fishing zones should be avoided due to restrictions?

USER FLOW



ARCHITECTURE & DATA FLOW



Anti-hallucination gate — Synthesizer cites only verified numerical telemetry · strict formula & source checks before every advisory

Engineered System Architecture & Deployed Stack

- **Frontend & Maritime UX:** Next.js 15, Tailwind CSS, Framer Motion, Leaflet.js
- **Agentic AI Framework:** LangGraph StateGraph — 5 parallel specialist nodes, anti-hallucination gate
- **Backend & Async Engine:** FastAPI, Python 3.11, Uvicorn async pipelines
- **High-Speed Inference:** Groq LPU (Llama-3.3-70B @ 800+ tok/s, sub-second latency)
- **Multilingual Voice:** Bhasini ULCA ASR + Groq Whisper (~1.04s) + Edge-TTS, 10 Indic languages
- **Geospatial & Vector DB:** Supabase — PostgreSQL 15 + PostGIS for MPA & IMBL geofencing
- **Telemetry Ingestion:** INCOIS ERDDAP, Open-Meteo, ISRO MOSDAC layer

1 · TECHNICAL FEASIBILITY — SPEED & MODULARITY

- **Sub-second latency:** Groq LPU serves Llama 3 70B at 800+ tok/s — streams faster than coastal 2G/3G links to prevent timeouts.
- **Stateful DAG architecture:** LangGraph isolates specialist nodes — failure in one telemetry node never cascades or crashes the session.
- **Why LangGraph over n8n:** Strict TypedDict immutability, cyclic DAG with async fan-out, <10ms in-process execution (vs 1500ms n8n webhook lag).

2 · OPERATIONAL VIABILITY — DATA PIPELINES

- **Multi-source telemetry fusion:** INCOIS ERDDAP (SST, wind) + Open-Meteo (wave height) at zero cost; all data timestamped Live vs. Model-Baseline.
- **MOSDAC (ISRO) integration:** Provider-agnostic ingestion layer ready for live feeds once institutional account is approved; simulated fallbacks in the meantime.
- **Dual-provider ASR pipeline:** Bhasini ULCA ASR is primary speech-to-text; Groq Whisper is automatic silent fallback; Web Speech API is final failsafe.

3 · MITIGATION OF CRITICAL RISKS

! RISK — AI hallucinations on safety alerts

Mitigation: Strict tool-output grounding — the Synthesizer cannot invent metrics. Safety and HSI are always computed via the enforced formulas.

! RISK — Government API downtime

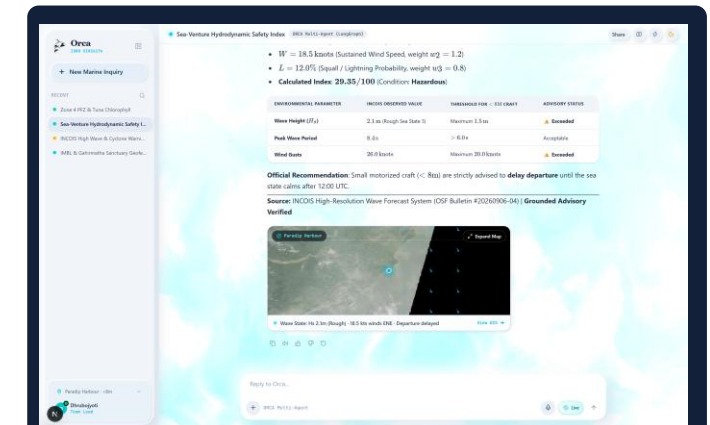
Mitigation: Supabase geospatial cache (30-min sliding TTL) serves the last verified state with an explicit “data valid as of [timestamp]” watermark.

! RISK — Official data with no public API

Mitigation: A dedicated Public Source Research Agent checks a strict allowlist (mausam.imd.gov.in, ndma.gov.in) — zero-fabrication guarantee.

! RISK — Single ASR provider outage

Mitigation: Dual-provider ASR (Bhasini primary, Groq Whisper automatic fallback) ensures zero-downtime voice input under adverse connectivity.



Live Sea-Venture Hydrodynamic Safety Index — grounded, cited advisory inside the ORCA UI (Risk #1 mitigation in action).

THE IMPACT LOOP



Fuel & time savings compound into direct household income, lower emissions, and safer, more resilient coastal communities.

ECONOMIC

- **Fuel & time optimization:** INCOIS PFZ heatmaps cut search time by 60–70% versus trial-and-error fishing.
- **Profit multiplier:** Vernacular heatmap delivery saves diesel per trip and lifts catch yield 2–4x.
- **Macro scale:** Directly boosts a sector contributing ~1.07% of India's GDP and the wider blue economy.

1.07%

of India's GDP — the blue economy ORCA strengthens

SOCIAL

- **Preventing loss of life:** Proactive spoken warnings (“2.5m waves — unsafe for craft <8m”) prevent capsizing in squalls.
- **Bridging the digital divide:** Vernacular voice queries democratize satellite data for zero-literacy fishers.
- **Zero-phone roadmap:** Toll-free IVR, USSD/SMS, coastal VHF radio, ISRO NavIC D2D & harbour kiosks reach offline fishers too.

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Indic languages supported end-to-end

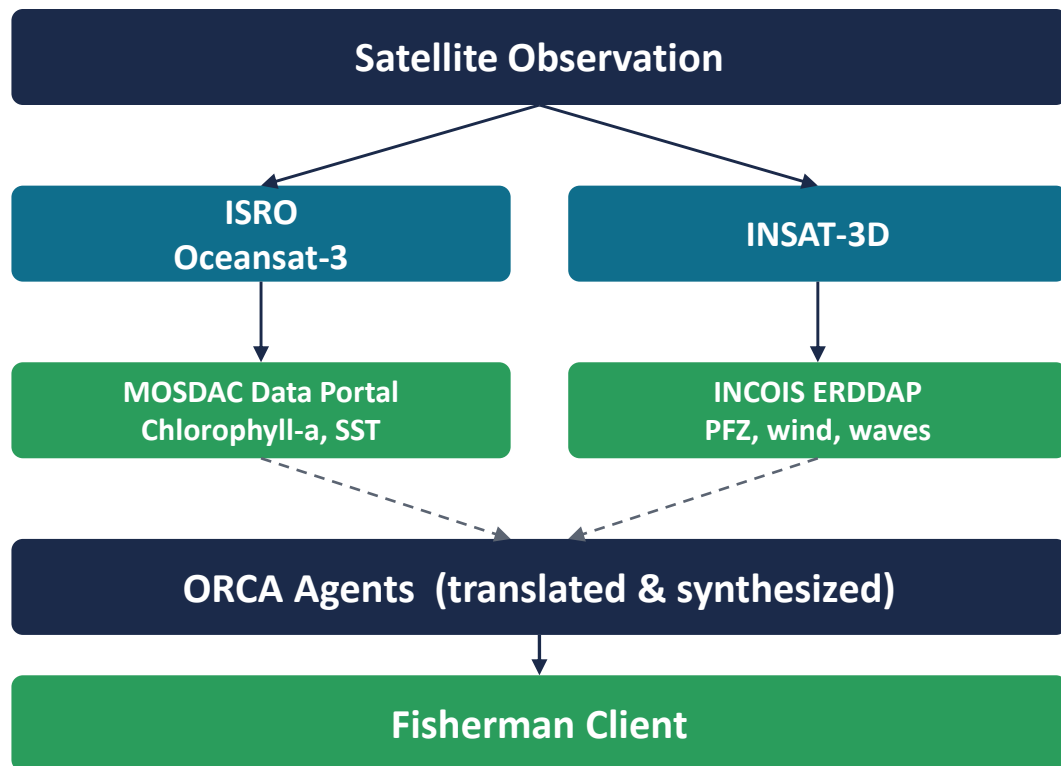
ENVIRONMENTAL & GEOPOLITICAL

- **Carbon footprint reduction:** Slashing search time removes millions of litres of burnt diesel from the atmosphere annually.
- **Dynamic geofencing:** The Risk Agent checks GPS against Marine Protected Areas and the IMBL, averting geopolitical arrests at sea.

28M

coastal lives within ORCA's reach

DATA VALIDATION FLOW



LIVE OPERATIONAL DEPLOYMENT (TRL-6)

- Web Application: <https://orca-seven-alpha.vercel.app>
- Backend API Docs: <https://orca-w0zz.onrender.com>
- Video Demo: <https://youtu.be/sYZFu3BxVJA>
- Codebase: <https://github.com/dhrubojyotihazra/ORCA>

VALIDATED SOURCES & IMPLEMENTATION

Domain	Validated Source / Reference	Implementation in ORCA
Economic Data	NCAER Report (Ministry of Earth Sciences) — “Estimating Benefits of Monsoon Mission & HPC”	Validates core assumption: PFZ advisories yield 60–70% fuel savings and ₹50,000 Cr national benefit.
Oceanography	ISRO MOSDAC (mosdac.gov.in) · INCOIS ERDDAP (incois.gov.in)	INCOIS ERDDAP is live source for SST & wind; MOSDAC layer pending institutional approval.
Fishery Biology	ICAR-CMFRI & INCOIS Marine Fishery Biology Envelopes (cmfri.org.in)	Defines Gaussian thermal & trophic envelopes for species-specific Habitat Suitability Index.
Agentic AI	LangGraph Framework (LangChain) — “Multi-Actor Stateful Applications with LLMs”	Backbone of ORCA's cyclic DAG architecture, enabling isolated reasoning & anti-hallucination.
LLM Inference	Groq LPU Architecture Whitepapers · Meta Llama 3.3 70B (wow.groq.com)	Delivers sub-second inference via Groq LPU (>800 tok/sec) to eliminate coastal latency.
Maritime GIS	Marine Regions (marineregions.org) — EEZ & Maritime Boundaries Datasets	Exact GeoJSON boundary layers used by PostGIS for real-time MPA & boundary geofencing.



SCAN IT TO SEE
THE PROTOTYPE !